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AND THE CATEGORIZATION OF MATHEMATICAL OBJECTS

ANTON DOCHTERMANN

ABSTRACT. In [2] Lakoff and Nuñez develop a basis for the cognitive science of *embodied mathematics*. For them, the abstract concepts and reasonings of mathematics are grounded in the conceptual processes that we develop in our interaction with the physical world. Through the use of conceptual metaphor these cognitive devices are projected to the realm of mathematical reasoning. In this paper we employ the methods spelled out in [2] to make a thorough study of the *equivalence relation*, a particular mathematical object. In particular we identify the primary cognitive metaphors at play in the conception and understanding of the equivalence relation and discuss some examples. The equivalence relation is a fundamental object in all branches of mathematics and, although a relatively simple construction, its cognitive analysis proves to be a rich source of cognitive metaphors and conceptual blending.

1. INTRODUCTION

Categorization is a process that plays an important role in our everyday lives. Much of human abstract thinking, and cognition in general, is involved in the process of categorizing. Be it objects, concepts, etc., we seem to make sense of the world by creating little boxes in which to place various divisions of it. The inclination to categorize is apparent at a very early age; it is a feature of the way that we interact in our daily lives with the objects that surround us. Categorization is part of our experience.

One of the earliest and historically most influential theories of human categorization dates back to the Classical Theory of Aristotle. The Classical Theory supposes that every category can be defined by a list of necessary and sufficient conditions. A particular mental object is in the category if and only if it fulfills these conditions. The members of a category, then, are precisely those objects that possess each of these conditions [1]. For Lakoff, the Classical Theory is an ‘expert version’ of a more general Folk Theory of Essences that influences much of Western thought. In that theory, it is assumed that every specific thing is a type of thing (i.e., can be categorized), and

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that the conditions that define a particular category are inherent properties that the members of the classes possess [2]. These assumptions are inherent in the Classical Theory of categorization.

Although more recent experiments and theories strongly suggest that the Classical Theory does not fit the way that we use classes in everyday language (see Appendix), along with the Folk Theory of Essences, it does influence that way that we conceptualize the notion of a categorization. In other words, it influences the way we think *about* categorization.

2. THE ‘COMPLETE’ CATEGORIZATIONS OF PHYSICAL OBJECTS

The cognitive process of abstract categorization is grounded in our experiences with categorizing physical objects in our spatio-temporal experience. Dogs, cats, tress, apples; these are the first categories that we learn. To understand the cognitive processes involved in categorizing mathematical objects, then, we must first investigate the structure of physical object categorizations.

As we have seen, the Folk Theory of Essences motivates the way that we conceptualize the process of categorization. Inherent in this theory is the assumption that ‘every specific thing is a kind of thing’ [2]. To borrow Lakoff’s example, there are kinds of animals: dogs, cats, birds, etc., and each specific animal is a kind of animal. This is a categorization of physical objects: animals from our real world experiences are categorized into a finite number of ‘different kinds’ of animals (dogs, cats, etc.). Again, the Folk Theory of Essences tells us that each specific instance of animal fits into one of our categories; this includes instances of animals that we may or may not encounter in our experience. Inherent in the cognitive interpretation of the categorization, then, is the conception of the *totality* of animals. In other words, our categorization is understood as a *complete categorization*.

This presumption is not trivial. In reality, our preconceived categories give us only the potential for a complete categorization, at best. It requires additional cognitive apparatus to conceptualize a particular categorization as complete. The notion of completeness is inherent in all of the cognitive categorizations that we make, and, as we will see, it plays a significant role in the cognitive classification of mathematical objects.

2.1. Finite Collection Categorization. In some ways, our notion of the completeness of a categorization is a consequence of the Folk Theory of Essences, but I would also argue that it comes as an entailment of a cognitive metaphor that we use to ground general categorizations. Our real world experiences with categorization involve finite collections of physical objects, and each of these experiences has a general structure to them: we are given (or construct) categories to place the objects, we place the objects accordingly, and we understand that the process is ‘done’ when all of the objects have been placed.

A child is given a bowl of fruit and is told to sort them into piles of apples, oranges, and pears. Inherent in this process of categorization is the understanding that each object in the bowl belongs to exactly one category of fruit (otherwise, we would not have given her the task). For the

child, categorization is complete when each fruit from the original collection has been placed in the appropriate pile. In the case of finite collections of this kind, categorization involves well-defined categories such that no object can belong to more than one category at any one time (again, if the categories were ambiguous, we would not have given her the task). We understand that when our categorization is complete, each object sits in exactly one category.

Much of our early education is occupied with these finite collection categorizations, and it is from these experiential schemas that we metaphorically conceptualize the notion of categorization in general. Here is the structure of that metaphor:

The Complete Categorization Metaphor

Source Domain	Target Domain
Categorizations of finite collections of objects	General object categorizations
Categories to place objects	Preconceived categories
Each object is placed in a category	Each thing is in a category
Entailment 1: Categorization is done when all objects are placed in an appropriate category	All things are categorized
Entailment 2: When the categorization is done, each object sits in exactly one category	Each thing is in exactly one category
Entailment 3: The collection of things given to categorize	There is a class of things categorized

3. THE CONFLATION OF CATEGORIZATION AND CLASSES

In our real world experiences with finite object collection categorizations, we are given a collection of objects to categorize. As we have seen, this process schema grounds our conception of general object categorizations. An entailment of this conceptual metaphor is the notion that our categories sit inside a class of objects to be completely categorized. In other words, the Complete Categorization Metaphor induces a conflation of cognitive processes: the cognitive activity of categorizing conflates with our conceptualization of a class.

The concept of a class, then, is intimately linked to the process of categorization in our everyday experience. Note that the class conception is not linked to the categories themselves; rather, it is the class of totality of objects that are classified. We will see that the conception of a class at this level is significant to the classification of mathematical objects. However, we must first investigate the cognitive operations involved in the conceptualization of a class.

3.1. Classes. As Lakoff argues, the concept of a class is experientially grounded via the ‘Classes are Containers’ metaphor. Let us emphasize here that by ‘class’ we do *not* mean the mathematical

notion of a collection of sets used in set theory, but rather the construction of the more general cognitive notion. We conceptualize classes in terms of container schemas: elements of a particular class are objects in a container. The interior of the container constitutes the elements of a class, the exterior the nonelements. We use these mental container schemas in ‘perceiving, conceptualizing, and reasoning about our everyday experience’ [2].

As we have seen, modern mathematics is an expression of the axiomatic method. A finite number of axioms and definitions define the objects within a particular branch of math, and the container schema metaphor is used to conceptualize the class of these objects. At their face value, however, a finite list of axioms and definitions cannot provide any sort of collection of objects. At best, the axioms and definitions give us a way to ‘check’ a particular object’s properties against a list of criteria.

We need an additional metaphor to cognitively understand the list of criteria as actually constituting a class of objects. Again, we turn to the Folk Theory of Essences. As we have seen, the Folk Theory tells us that every specific thing is a kind of thing. In addition, it assumes that each thing has a collection of properties that makes it the kind of thing that it is, and that these ‘essences’ are in some sense *causal* they are inherent to that thing. The Folk Theory of Essences motivates the Classes are Properties Metaphor:

The Classes are Properties Metaphor

Source Domain	Target Domain
A List of Properties and the Folk Theory of Essences	Classes
Finite list of properties	Class C of objects with those properties
Object that has all of the properties	Object in the class C
Object that does not have all of the properties	Object not in the class C
All objects have a list of properties that make them what they are	All objects are in some class
We can compare our list with any other list	All objects are in the class C or not in the class C

Sometimes, a class of mathematical objects comes to us in a natural way (i.e., the integers and rational numbers), and sometimes, given a list of axioms and definitions, we have to construct the classes ourselves based on the desired properties via the Classes are Properties metaphor.

3.2. Sets. Only in the last century has the notion of a class of mathematical objects become a focus of rigorous study. The axioms of set theory that describe formal classes of mathematical objects are good indications of the cognitive metaphors inherent in the conception of these classes.

First, we need ‘Boole’s Metaphor’ [2] to cognitively map an algebraic structure onto the domain of classes (where the operations are union and intersection) to give us the well-known notions of an *empty class* and a *subclass*. Next, we insist that sets can be members of other sets; that is, we must conceive of them as objects in the same way that we understand elements of a class as objects. The Sets are Objects metaphor [2] is fundamental to the cognitive conception of mathematical sets, and, as we will see, to the application of the equivalence relation as well.

4. EXPERIENCE WITH OBJECTS THE OBJECT DECONSTRUCTION SCHEMA AND THE PARTITION ENTAILMENT

Object construction and deconstruction is an experientially derived schema that grounds some of our most common conceptual metaphors (including the cognitive conception of number [2]). The deconstruction of an object is the process of physically ‘splitting up’ an object into a finite number of components. Like the slices of a loaf of bread, our experience tells us that these components are physically independent of one another (are disjoint), and that the collection of components (slices) taken as a whole constitutes the entire object (the loaf).

Now, as we have seen, a feature of the conceptual blend that describes our cognitive understanding of a set of mathematical objects is the metaphor that sets are objects. This metaphor, blended with the object deconstruction/construction schema, provides us with the cognitive understanding of a ‘partition’ of a set. In first order logical terms, a partition P of a set S is a set of nonempty subsets of A that are disjoint and exhaustive; i.e., no two different sets in P have any common elements, and each element of A is in some set of P . This definition makes sense in terms of our grounding metaphors for a set. If a set is an object in our spatial-temporal experience, then, we have the ability to ‘cut’ it up into disjoint pieces like the slices of a loaf of bread. Again, we understand that each ‘slice’ is physically independent (disjoint), and also that the collection of slices constitute the whole set.

The Sets are Objects Metaphor and the Partition Entailment

Source Domain	Target Domain
Objects	Sets
Objects can be cut up into parts	Sets can be partitioned
A physical slice of the object	A nonempty subset, constituting a member of the partition
Each slice is physically independent	Each subset is disjoint (do not share elements)
The totality of the slices constitute the whole object	The union of the subsets is the set

A typical way to visualize partitions of sets is to draw the set as a circle and to divided the space up into boxes:

These representations are a good indication that the inherent Sets are Containers metaphor is also playing a role in our cognitive understanding of a partition. It is interesting to note, however, that the physical location of the ‘slices’ in this representation is not an entailment; we can partition a set in ways that may not coincide with our graphical notation of it. It is also worth noting that we are not restricted to finite sets or even finite partitions; grounded in the domain of finite physical containers/objects, we can use the Basic Metaphor of Infinity [2] to conceptualize a partition with an infinite number of elements.

5. THE ‘CATEGORIZATION OF A SET IS A PARTITION METAPHOR’

As we have seen, the cognitive process of a categorization brings with it the conceptualization of a class of objects to be categorized. Also, we saw that our understanding of a categorization is grounded in our experience with finite collections of objects through the Complete Categorization metaphor. Entailment 2 of this metaphor grounded our notion that, in a categorization, a particular object is in exactly one category. This means that the categories are disjoint in the same way that a partition of a set provides a set of disjoint subsets; remember that this came as an entailment of the Sets are Objects Metaphor. But, cognitively speaking, a mathematical set is also grounded in the same container schemas that provide us with classes. In other words, the cognitive conception of a categorization and that of a mathematical set are linked to the same grounding schemas. At the same time, we have a property of one conception that correlates to property of the other (namely, the disjointness of partitions and of categories). Again, there is a conflation of cognitive processes that results in the Categorization of a Set is a Partition metaphor cognitively, we begin to think of the categorization of a set as a partition.

5.1. Partitions and Relations. In most cases, a certain collection of mathematical objects that we wish to investigate forms an infinite set, and even a coherent cognitive understanding of a partition does not provide us with sufficient cognitive apparatus to construct a particular partition of this set. As we have seen, the axioms and definitions of a particular branch of math provides us with an infinite collection of mathematical objects (under the metaphor that Classes are Properties). The goal will be to categorize these objects in some meaningful way by constructing a partition of the set (using the metaphor that Categorization of a Set is a Partition). Given an infinite set, this is a difficult thing to formalize.

Suppose instead that we start with a given partition on a set A (take A to be the set of Natural numbers, let our partition consist of the set of even numbers and the set of odd numbers). Conceptually, we have taken the class of Natural numbers and cut it up so that each element is in exactly one of two ‘little boxes’ (subsets). Now, suppose that we define a binary relation R on A as follows: two elements (numbers) x and y are in the relation R (we will write xRy) if and only if they are in the same ‘little box’. Immediately, we see that the relation R will have the following three properties:

R is Reflexive: for all x in A , xRx

R is Symmetric: if xRy then yRx

R is Transitive: if xRy and yRz then xRz

5.2. The Equivalence Relation Definition and Properties. We call R an equivalence relation on a set A if R is a binary relation on A that is reflexive, symmetric, and transitive (Enderton, 1977). Note that the understanding of a set is inherent in this definition; the cognitive conception of the equivalence relation, then, is grounded in our conception of a set (that is why we spent the time investigating sets).

Now, in the effort to minimize our axioms and definitions, we understand that the three properties of the equivalence relation are in fact independent; if one of them was derivable from the other two, then we could not define a binary relation on a set that had the other two but not it. And perhaps the best way to understand each of the three properties and their consequences is to examine relations that have exactly one of them lacking:

A Reflexive and Symmetric, but not Transitive relation: consider the set of real numbers, with the relation ‘having an absolute difference of 1 or less’. Now, the integers 2 and 3 are in relation, as are 3 and 4, and yet 2 and 4, having a difference of 2 are NOT in relation. In real analysis (the basis of calculus), the absolute difference is considered a metric on the set of real numbers in that it fulfills the properties that we have axiomatized for a metric space. Having a difference of 1 or less, then, is a metaphorical notion of ‘closeness’ within the realm of the set of real numbers. It does not define an equivalence relation, however, because it is not transitive.

A Reflexive and Transitive, but not Symmetric relation: consider the set of integers, with the relation ‘is less than or equal to’. Now, 2 is in relation to 3, but 3 is NOT in relation to 2. In this example, we can see that the anti-symmetric property ‘lines’ up the set of distinct integers in increasing order. In fact, a binary relation that is antisymmetric and transitive is called ‘linear ordering’, for good reason (Enderton, 1977).

A Symmetric and Transitive, but not Reflexive relation: consider the set of integers, with the relation ‘having the same natural log value’. Since the log function is not defined for negative numbers, the relation is not reflexive. The reflexive property guarantees that the relation holds on the entire set, and without it, we can not hope to form a partition of a set because some elements will be missed.

5.3. Equivalence Classes and Partitions. We have shown how a partition on a set A can induce an equivalence relation (the relation is simply ‘in the same box’). But, again, our goal is to formalize a process of constructing a partition on a given set. It turns out that, under certain conceptual metaphors, any relation on a set that is an equivalence relation induces a unique partition on a set. In other words, the categorization of a set of mathematical objects is complete (metaphorically, of course) when we define a relation on its that is reflexive, symmetric, and transitive.

Suppose that we have an equivalence relation R on a set S. For a particular element x of S, we define the ‘equivalence class of x’ (denoted $[x]$) as the set of elements in S that are in the relation R with x. Next, take the collection E of all such equivalence classes. I claim that E is a partition of the set (in the sense defined above). Because R is reflexive, we have that each

equivalence class is nonempty (the class $[x]$ contains at least the element x) and is a subset of A (because R is a binary relation on A). The main thing to check is that the collection of equivalence classes is disjoint. The mathematical proof utilizes the symmetry and transitivity of the relation R . Suppose on the contrary that $[x]$ and $[y]$ had some common element t . This means that xRt and yRt (where, again, xRt denotes that x is in relation to t). By symmetry on yRt , this implies tRy , and by transitivity, we have xRy . Now, let s be any element of $[y]$. We have then yRs , which implies xRs by transitivity. This means that s is in $[x]$ (by definition), which implies that $[y]$ is contained in $[x]$ (since s was an arbitrary element of $[y]$). Since R is symmetric, we also have yRx and we can reverse this last argument to conclude that $[x]$ is contained in $[y]$. Finally, this gives us that $[x] = [y]$; i.e., whenever two equivalence classes share a common element, they must on fact be the same equivalence class.

5.4. The ‘Partitions are Defining Equivalence Relations’ Metaphor. According to the mathematical proof outlined above, any equivalence relation R on a set S induces a unique partition of that set. If we are to investigate the structure of the proof, we see that it involves quantification over the set S ; given any element x in our set S , we can show that it is in exactly one equivalence class. Because this element was chosen arbitrarily, we conclude that each element of S is in precisely one equivalence class; thus the collection of equivalence classes forms a partition. Now, in the application of equivalence classes to a particular mathematical domain of objects, it will be essential to think of a particular category (equivalence class) as a unique mathematical object itself within a collection of class-objects. In other words, we need to have a conceptualization of the entire class and of the entire collection.

In the case of finite sets, we can apply a particular equivalence relation to a set of objects to produce a complete partition in a relatively straightforward way: we pick the first element out and place it in its own equivalence class; subsequent elements are classed according to their relationship with chosen elements. At some point, the process is done and we have a partition of the set. Typically, however, equivalence relations are defined on infinite sets, and the best we can do is to show that, given any element x , x belongs to exactly one equivalence class. We need extra cognitive apparatus to understand that this process actually gives us a complete partition with complete object level classes. We need to apply the relation to all elements simultaneously, and run each of them through the proof outline above, all at the same time, keeping track of which elements went where and what each equivalence class looks like. It is a conceptual metaphor that identifies our notion of an equivalence relation with that of a partition.

The Partition is Defining an Equivalence Relation Metaphor

Source Domain	Target Domain
Defining an Equivalence Relation	Partitions
Set S of elements the relation is defined on	Set S of elements to be partitioned
Two elements x and y are in a relation R	Two elements x and y are in the same subset
Two elements y and z are not in relation R	Two elements y and z are not in the same subset
Given any element of S , we can show that it belongs to exactly one equivalence class	The equivalence classes are nonempty, disjoint, and comprise the whole set
Entailment 1: Defining an equivalence relation on S	Partitioning the set S
Entailment 2: The structure of the equivalence relation	The structure of the partition

This metaphor is central to the cognitive understanding of the equivalence relation and its application to sets of mathematical objects. The real power of the metaphor lies in the fact that we are provided with a well-defined and discrete process of forming partitions of infinite sets of mathematical objects. As we have seen, partitions of sets are conceptualized as categorizations, and so, in essence, we now have a way to categorize mathematical objects.

As an entailment of this metaphor, the cognitive understanding of the structure of a particular partition becomes an understanding of the equivalence relation that induces it. In our experiences with finite object collections, a partition does not require us to have any a priori knowledge of the elements themselves, let alone the relationship between any two given elements. However, correlation in our experience between the two domains motivate their metaphorical relationship.

5.5. A Partition is a Set. At this point, it might be instructive to consider the set theoretical justification of a partition of a set A . First, for any element x of A , we have the equivalence class $[x]$ is a set by a subset axiom of set theory that guarantees the existence of subsets described by certain first order logical predications. Furthermore, the collection of equivalence classes forms a set because they form a subset of $P(A)$, the power set of A (the set of all subsets of A); and by the power set axiom, we have that $P(A)$ is a set.

6. 'EQUIVALENCE CLASSES ARE OBJECTS' METAPHOR

As we have seen, our notion of a set is grounded in our Boolean class conceptualization along with the additional Sets are Objects metaphor. In our applications of equivalence relation categorizations, as well as in set theoretical terms, equivalence classes are sets. The Equivalence Classes are Objects metaphor, then, is in some ways a special case of the Sets are Objects metaphor.

The equivalence classes that are given to us by our particular relation are not collections of objects, but rather mathematical objects themselves with ‘object-level’ properties.

6.1. Equivalence Classes are Objects.

7. INSTANCES OF EQUIVALENCE RELATIONS GROUNDING THE RELATIONS

Formal mathematics is concerned with investigating sets of mathematical objects, and, as we have seen, those sets sometimes come naturally and are sometimes created metaphorically by a list of properties. We are inclined to categorize these objects in specific ways, but the question remains: what motivates these particular categorizations? We have seen how an equivalence relation is the metaphorical means of categorizing a set of mathematical objects by the partition that it induces. The motivation behind a categorization, then, lies in the structure of the equivalence relation itself.

Now, the cognitive advantage of categorizing a particular set of objects is that we can use the Sets are Objects metaphor to treat the category itself as a single mathematical object. Within a particular domain of mathematics, all elements of the category are identified as instances of the ‘same’ object. The equivalence relation that induces the categorization of a set, then, is a characterization of the conceptual metaphors that govern our understanding of that set. Equivalent elements of a set (members of a category, of an equivalence class) are determined by the metaphors that ground our cognitive conceptualization of them. The cognitive conception of a categorization is the cognitive justification of the equivalence relation.

Perhaps the best way to illustrate this process is to investigate the cognitive processes involved in some of the more fundamental instances of equivalence relations and categorizations. Although equivalence relations pervade nearly all branches of formal mathematics, many relations are not particular to ‘college level’ math; they are inherent in the way we understand some of the most basic mathematical objects.

7.1. Rational Numbers. At an early stage in our mathematical education, we are told that the numerals ‘2/3’ and ‘4/6’ designate the ‘same’ rational number; that in fact any given fraction can be represented in a multitude of ways (6/9, 8/12, etc., are other candidates). We are given a general rule: two rational numbers, a/b and c/d , are the same if and only if they satisfy the equation: $a * d = b * c$. Here, we have made precise the rather abstruse notion of equality of fractions in terms of a more familiar multiplication of integers. This is a rather deep and subtle notion, and the fact that we can make the move at such an early age illustrates some of our most basic cognitive handlings of mathematical objects.

Now, we can easily see that the relation on the set of integers as described above is a binary relation that is reflexive, symmetric, and transitive. Thus, it is an equivalence relation and we have a metaphorical partition on the set of ratios of integers. As we have said all along, the power of the equivalence relation is that it allows us to think of the classes (categories) as the mathematical objects. This is clear in the case of the rational numbers. The object 2/3, while simultaneously representing 4/6, 6/9, etc., is cognitively understood as a single number, not as a collection of

numbers or ratios. Each equivalence class is a rational number and the set of equivalence classes is the set of rational numbers.

When we say that ' $2/3 = 4/6$ ' as rational numbers, what we are really implicitly understanding is that, for any two rational numbers a/b and c/d , we have defined $a/b = c/d$ if and only if $ad=bc$ (notice that we understand $ad=bc$ as equal result frames stating that the resulting number obtained from the two products is the same). In a strictly cognitive sense, our understanding of $2/3$ is distinct from that of $4/6$, and we need extra cognitive apparatus to comprehend them as 'equal'.

As we have seen, the metaphors that ground equivalence relation categorizations tell us that the structure of the categorization is described by the structure of the equivalence relation. In many cases, the way that we define an equivalence relation on a class of mathematical objects is motivated by the cognitive blending of the conceptual metaphors that we use to ground the objects in our everyday experience. The equivalence relation that gives us the rational numbers is a good example of this.

As Lakoff argues, our conception of the rational numbers can be understood as a blend of four distinct metaphors grounded in experiential schemas: object collection, object construction, measuring sticks, and motion along a line. Within the domain of object construction, the notion of $2/3$ is much different than that of $4/6$ - in the one case we imagine a collection of two pieces, in the other we have four. However, if we are to conceive of these numbers as points along a measured line, we can indeed think of them as representing the same value.

7.2. Bijections. A bijection from one set to another is a one-to-one and onto correspondence. If we are to consider a set S of collections, and define a relation 'in bijection with', then we see immediately that:

- (1) every collection is in bijection with itself (just take the identity mapping)
- (2) if a collection A is in bijection with B , then B is in bijection with A (every bijection has a bijective inverse)
- (3) if a collection A is in a bijection b_1 with B , and B is in bijection b_2 with C , then A is in bijection with C (consider b_2 composed with b_1)

In other words, 'in bijection with' is an equivalence relation. The bijection correspondence is used in many branches of math.

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